

terms, equating a password protected file to a safe, there are multiple points of vulnerability. It is not enough to make the key too complicated to duplicate if the safe itself is weak; it is not enough to have a complicated password if the encryption algorithm itself is weak. It is important that the safe not have any design flaws which enable someone to subvert the key mechanism. If all else fails, you can simply drill through the safe lock; at this point, there is nothing that can be done to protect the safe as this is equitable to a brute-force attack – in fact this is where the term comes from.

The password algorithm is not the only thing standing in your way either. If you are trying to recover the password for a remote computer or for a system using a authentication mechanism, you need also be aware that the system could limit the number of tries you get to guess the password. For example if the system blocks access to an account after 5 failed password entry attempts, you cannot use a brute force or even dictionary attack.

What if however you simply find the safe key lying written on a piece of paper in a drawer near the safe? Or if it is a physical key, what if it is lying hidden somewhere in the room?

This is where social engineering kicks in, where instead of hacking away directly, we try first to get as much information as possible about the person being hacked in order to make more reasonable guesses. The art of gathering such information and making use of it, is called Social Engineering.

2.1.2 Social Engineering

“There is no patch for human stupidity”

The simplest way to describe Social Engineering would be that it is a means of gathering information about a target using social means rather than purely electronic means. Social engineering works by exploiting bugs and vulnerabilities of the human mind instead of just those of computer systems. A “social engineer” tries to get the target to divulge as much information about them as possible in order to improve their chances at guessing the victim’s password.

Actually, such a technique could be used for more than just guessing passwords. Con artists often rely on learning about their victim in order. A non-password-guessing usage would be to con a person by impersonating as a family member or a friend. Let us look at an example:

A con artist gains as much information as possible about a child in a school, finds out a child’s friends’ names, their parents’ names, the teachers name, and as much more information as possible. This person then visits the parents of said child, pretending to be someone of authority from the school, asking for payment for a new initiative by the school.

This oversimplified example perhaps has many vulnerabilities in itself, however it does illustrate a point. Social engineering is one of the biggest